LiAL. - temă seminar 5

1. $U = \{(x, y, z) \in \mathbb{R}^3 \,/\, -x + 3y + z = 0\} \subset \mathbb{R}^3/\mathbb{R}$

a) fie $v_1, v_2 \in U$ $\Rightarrow$ $v_1 = (x_1, y_1, z_1)$, $-x_1 + 3y_1 + z_1 = 0$

$\alpha_1, \alpha_2 \in \mathbb{R}$ $\quad v_2 = (x_2, y_2, z_2)$, $-x_2 + 3y_2 + z_2 = 0$

$\alpha_1 v_1 + \alpha_2 v_2 \in U$?

$\cancel{\alpha_1 v_1 + \alpha_2 v_2 = (-\alpha_1 x_1 - \alpha_2 x_2, 3\alpha_1 y_1 + 3\alpha_2 y_2, \alpha_1 z_1 + \alpha_2 z_2)}$

$\alpha_1 v_1 + \alpha_2 v_2 = (\underbrace{\alpha_1 x_1 + \alpha_2 x_2}_{x}, \underbrace{\alpha_1 y_1 + \alpha_2 y_2}_{y}, \underbrace{\alpha_1 z_1 + \alpha_2 z_2}_{z})$

$-x + 3y + z = -\alpha_1 x_1 - \alpha_2 x_2 + 3\alpha_1 y_1 + 3\alpha_2 y_2 + \alpha_1 z_1 + \alpha_2 z_2$

$= \alpha_1 (\underbrace{-x_1 + 3y_1 + z_1}_{0}) + \alpha_2 (\underbrace{-x_2 + 3y_2 + z_2}_{0})$

$= 0 \Rightarrow \alpha_1 v_1 + \alpha_2 v_2 \in U \Rightarrow U \subset \mathbb{R}^3$ sp vect

b) fie $v \in U$ vector arbitrar, $v = (x, y, z)$, $-x + 3y + z = 0$

$x = 3y + z$

$U \ni v = (3y + z, y, z) = (3y, y, 0) + (z, 0, z)$

$= y \underbrace{(3, 1, 0)}_{u_1} + z \underbrace{(1, 0, 1)}_{u_2} = y u_1 + z u_2 \Rightarrow$

$S' = \{u_1, u_2\} \subset U$ sis de gen (1)

fie $\alpha_1, \alpha_2 \in \mathbb{R}$ a.î. $\alpha_1 u_1 + \alpha_2 u_2 = 0_{\mathbb{R}^3}$

$\alpha_1 (3, 1, 0) + \alpha_2 (1, 0, 1) = (0, 0, 0)$

$\Rightarrow \begin{cases} 3\alpha_1 + \alpha_2 = 0 \\ \alpha_2 = 0 \\ \alpha_1 = 0 \end{cases} \Rightarrow \alpha_1 = 0, \alpha_2 = 0$ sol unică $\Rightarrow$

$\Rightarrow S' = \{u_1, u_2\} \subset \mathbb{R}^3$ s.v. lin indep (2)

$\overset{(1),(2)}{\Rightarrow} \underset{\text{bază}}{S'} \subset U \Rightarrow \dim_{\mathbb{R}} U = 2$